Julius Alexander Zeiss, M.Sc. Physics

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Profile

- I develop classical algorithms that make difficult quantum problems tractable. Combining group theory, quantum Shannon theory, and convex optimization, I use quantum de Finetti theorems and symmetry-reduced semidefinite programming hierarchies to obtain rigorous bounds on quantum correlations and quantum error correction. Alongside these algorithmic questions, I study the power and limitations of entanglement in communication tasks and non-local games, especially quantum graph coloring. My work connects quantum information with graph theory and combinatorics, investigating how entanglement changes classical constraints and how graph structure shapes quantum strategies.
- Speaker at leading international conferences (TQC, Beyond IID, AQIS) and invited speaker at institutions including Stanford University and the Perimeter Institute for Theoretical Physics.
- Founder and host of RWTH Aachen's regular A.I. Workshop on rigorous, reproducible science with A.I.

Education

since 03/2023	**PhD in Physics (Quantum Information Theory), RWTH Aachen University, Germany** Institute for Quantum Information, final-year PhD student Expected completion: 31 March 2027 Advisor: Prof. Dr. Mario Berta
10/2019 – 08/2022	**M.Sc. Physics, University of Cologne & University of Bonn, Germany** Focus: theoretical and mathematical physics Thesis: *SDP Relaxations for Quantum Inflation* Advisor: Prof. Dr. David Gross, Thesis grade: 1.0 (very good) Overall grade: 1.4 (very good)
10/2018 – 08/2019	**Master's studies in Physics, University of Münster, Germany** Transferred to Cologne/Bonn Focus: theoretical physics, quantum information theory, biophysics
10/2015 – 09/2018	**B.Sc. Physics, University of Münster, Germany** Thesis: *Numerical Treatment of the Manning Potential for the Description of Ammonia* Advisor: Prof. Dr. Gernot Münster, Thesis grade: 1.0 (very good) Overall grade: 1.8 (good)
04/2013 – 09/2015	**Bachelor's studies in Business Administration, University of Münster, Germany** Transferred to physics
09/2010 – 05/2012	**International Baccalaureate, King William's College, Isle of Man** Overall grade: 41/45 points (very good, 1.1 German Abitur)

Publications and Preprints

2026	**J. A. Zeiss**, *Hedetniemi's conjecture for Shannon OR-capacity: counterexamples and asymptotic separation*, arXiv:2610.41977
2026	**J. A. Zeiss**, *Comment on "The polarization hierarchy for polynomial optimization over convex bodies, with applications to nonnegative matrix rank"*, arXiv:2610.24542 (submitted to Linear Algebra and its Applications)
2026	**J. A. Zeiss**, *A counterexample to the quantum Hedetniemi conjecture*, arXiv:2609.20690
2026	**J. A. Zeiss**, *Optimal entanglement-assisted source coding under a balanced-difference promise*, arXiv:2609.15757 • Lean formalization

2026 M. Berta, P. Costa Rico, G. Koßmann, L. Lami, **J. A. Zeiss**, *Sharp continuity of quantum conditional entropy*, arXiv:2607.24687 (under review at Annales Henri Poincaré)

2026 G. Koßmann, **J. A. Zeiss**, *Fixed points in de Finetti hierarchies*, arXiv:2607.23689 (under review at Communications in Mathematical Physics)

2026 **J. A. Zeiss**, G. Koßmann, R. Schwonnek, M. Plávala, *Finite de Finetti for convex bodies and polynomial optimization*, arXiv:2601.15184 (under review at Mathematical Programming A)

2025 **J. A. Zeiss**, G. Koßmann, O. Fawzi, M. Berta, *Approximating fixed size quantum correlations in polynomial time*, arXiv:2507.12302 (under review at PRX Quantum)

2025 G. Koßmann, **J. A. Zeiss**, O. Fawzi, M. Berta, *On approximate quantum error correction for symmetric noise*, arXiv:2507.12326 (under review at Quantum)

Invited Talks

06/2026 **Stanford University, CS Theory Group, Palo Alto, USA**
A Family of Information-theoretic de Finetti Theorems for Constrained Optimization

04/2026 **University of Cologne, Quantum Information Seminar, Germany**
Approximating fixed size quantum correlations in polynomial time

07/2023 **Siegen University, Germany**
Inner approximation schemes for approximate quantum error correction

02/2023 **Perimeter Institute for Theoretical Physics, Waterloo, Canada**
Infinite Dimensional Optimisation Problems in Quantum Information — An operator algebra approach to the NPA Hierarchy

05/2022 **Alpen-Adria-Universität Klagenfurt, Austria**
SDP methods for the quantum causal compatibility problem

Contributed Talks

09/2026 **TQC 2026, Sherbrooke, Canada**
A Family of Information-Theoretic de Finetti Theorems for Constrained Optimization

05/2026 **ILAS 2026, Virginia Tech, Blacksburg, USA**
Convergent Inner-Outer Approximation Schemes From De Finetti Theorems For Games And Quantum Error Correction

05/2026 **ILAS 2026, Virginia Tech, Blacksburg, USA**
Information-theoretic finite de Finetti theorems for quantum and beyond[*]

03/2026 **Quantum at the Dunes, IIP-UFRN, Natal, Brazil**
Approximating fixed size quantum correlations in polynomial time

08/2025 **AQIS 2025, The University of Hong Kong, Hong Kong**
Approximating fixed size quantum correlations in polynomial time

07/2025 **13th Beyond IID in Information Theory, TUM Institute for Advanced Study, Garching, Germany**
Approximating fixed size quantum correlations in polynomial time

07/2025 **13th Beyond IID in Information Theory, TUM Institute for Advanced Study, Garching, Germany**
On approximate quantum error correction for symmetric noise[*]

[*] Talk given by a co-author.

Posters

04/2026 **851. WE-Heraeus-Seminar: 90 Years of Einstein-Podolsky-Rosen Paradox, Bad Honnef, Germany**
Approximating fixed size quantum correlations in polynomial time

01/2026 **QIP 2026, Riga, Latvia**
Approximating fixed size quantum correlations in polynomial time

08/2025 **QMATH Masterclass on Representation Theory in Quantum Information Science, University of Copenhagen, Denmark**
Schur–Weyl Duality for Efficient Approximation of Constrained Separability

07/2024 **Beyond IID in Information Theory, University of Illinois Urbana-Champaign, USA**
Approximating fixed size quantum correlations in polynomial time

Teaching and Supervision

since 03/2023 **RWTH Aachen University, Germany**
Head tutor and organizer: Advanced Seminar on Bell Inequalities (winter semester 2026/27)
Supervised two master's and two bachelor's theses in quantum information theory
Tutorials: Quantum Information Theory (master's; 2024, 2025), Quantum Mechanics (bachelor's; 2025), Statistical Physics (bachelor's; 2023, 2024)
Master's seminar: Advanced Topics in Quantum Information Theory (2024, 2026)

04/2021 – 04/2022 **University of Cologne, Germany**
Tutorials: Linear Algebra & Analysis, Proofs in Mathematics, Theoretical Physics

04/2019 – 09/2019 **University of Münster, Germany**
Tutorials: Applied Physics

Academic Service

Conference reviewing TQC (2023, 2024, 2025) • ISIT (2023, 2024, 2025) • QIP (2024, 2025, 2026, 2027) • Beyond IID (2023, 2024)

Journal refereeing IEEE Transactions on Information Theory (2025)

Outreach

since 2026 **A.I. Workshop, RWTH Aachen University, Germany**
Founder and host of a regular workshop on A.I. tools for research (from coding assistants and version control to formally verified proofs) and on using them to conduct better science

2024 – 2026 **ML4Q — Matter and Light for Quantum Computing, Germany**
Instructor: school internships (July 2024), Quanteen Day 2024, Quanteen Day 2026

11/2022 **DPG Physik Journal**
A. Matthies and J. Zeiss, *Quantum Computing*, review article on the Quantum Computing Bad Honnef Physics School, Physik Journal 21 (2022) No. 11, p. 57

Skills

Languages German (native), English (bilingual)

Coding Python, Mathematica, Lean (formal verification), Git, GitHub, Fortran, PHP, JavaScript, HTML, SQL, VBA

Other Professional Experience

2014, 2015 **Fielmann AG, Hamburg, Germany**
DCF modeling with Monte Carlo simulations, VBA and SQL solutions

01/2013 – 02/2013 **HMT, Hamburg, Germany**
Mathematical modeling using Monte Carlo simulations

08/2011 **AKO Capital, London, England**
Internship in financial research using mathematical modeling